

Customer Shopping Behavior Analysis

Data Analysis Report

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Tools Used: Python, SQL Server (SSMS), Power BI, Pandas, Matplotlib

1. Executive Summary

This project analyzes customer shopping behavior to understand purchasing trends, customer preferences, revenue generation, discount effectiveness, subscription patterns, and product performance.

The analysis combines Python for data cleaning and preprocessing, SQL for business-oriented querying, and Power BI for interactive dashboard visualization. The insights generated can help retailers improve marketing strategies, optimize inventory, personalize customer experiences, and increase revenue.

2. Project Objectives

The primary objectives of this project were to:

- Analyze customer purchasing behavior.
- Understand revenue generated across customer segments.
- Evaluate the impact of discounts on purchasing.
- Compare spending habits of subscribers and non-subscribers.
- Identify highly rated and frequently purchased products.
- Study customer loyalty based on previous purchases.
- Build an interactive dashboard for business decision-making.

3. Dataset Overview

The dataset contains 3,900 customer purchase records with 18 attributes describing customer demographics, purchasing behavior, payment preferences, shipping methods, discounts, subscription status, and product reviews.

Important Features

- Customer ID
- Age

- Gender
- Item Purchased
- Category
- Purchase Amount
- Review Rating
- Shipping Type
- Discount Applied
- Subscription Status
- Previous Purchases
- Frequency of Purchases
- Season
- Payment Method
- Location

4. Data Preprocessing (Python)

Several preprocessing steps were performed before analysis.

Data Cleaning

- Loaded dataset using Pandas.
- Checked data structure using info().
- Generated descriptive statistics.
- Checked missing values.
- Filled missing Review Ratings using the median review rating of each product category.
- Converted column names into lowercase.
- Replaced spaces with underscores.
- Standardized data formats.

Feature Engineering

Created additional business-friendly columns:

Age Group

Customers were segmented into:

- Young Adult
- Adult
- Middle-aged
- Senior

using quantile-based categorization.

Purchase Frequency (Days)

Converted purchase frequency into numerical values.

Example:

Frequency	Days
Weekly	7
Fortnightly	14
Monthly	30

Removed Redundant Features

Verified that

- Discount Applied
- Promo Code Used

contained identical information.

Therefore, Promo Code Used was removed to eliminate redundancy.

5. Database Implementation

The cleaned dataset was imported into Microsoft SQL Server.

A relational database named

customer_behavior

was created for performing SQL-based business analysis.

6. SQL Business Analysis

Several analytical SQL queries were executed to answer important business questions.

Revenue by Gender

Calculated total revenue generated by:

- Male customers

- Female customers

Purpose : Understand purchasing contribution across genders.

High-Spending Discount Customers

Identified customers who:

- Received discounts
- Still spent more than the average purchase amount

Business Value:

These customers represent premium buyers who respond well to promotional campaigns.

Top Rated Products

Retrieved the Top 5 products based on average customer review ratings.

Business Value: Helpful for inventory prioritization and promotional recommendations.

Shipping Analysis

Compared average purchase amount between:

- Standard Shipping
- Express Shipping

Business Value: Determines whether premium shipping customers spend more.

Subscriber vs Non-Subscriber Analysis

Compared:

- Number of customers
- Average purchase amount
- Total revenue

Business Value: Measures the effectiveness of subscription programs.

Discount Effectiveness

Calculated products having the highest percentage of discounted purchases.

Business Value: Helps identify products that rely heavily on promotional pricing.

Customer Segmentation

Customers were classified into:

Segment	Criteria
New	1 Previous Purchase
Returning	2–10 Previous Purchases
Loyal	More than 10 Previous Purchases

Business Value : Supports targeted marketing strategies.

Top Products within Each Category

Ranked the Top 3 purchased products in every product category using SQL window functions.

Business Value : Helps optimize inventory management.

Repeat Buyer Subscription Analysis

Studied whether customers with more than five previous purchases are likely to subscribe.

Business Value : Measures customer loyalty and subscription adoption.

Revenue by Age Group

Calculated total revenue generated by each age segment.

Business Value : Identifies the most profitable customer demographics.

7. Power BI Dashboard

An interactive Power BI dashboard was developed to visualize key business metrics.

Dashboard features include:

- Revenue Overview
- Customer Distribution
- Sales by Gender
- Revenue by Age Group
- Product Category Performance
- Subscription Analysis
- Shipping Analysis

- Discount Analysis
- Customer Segmentation
- Product Ratings
- Interactive Filters and Slicers

The dashboard enables stakeholders to explore customer behavior dynamically and supports data-driven decision-making.

8. Key Insights

The analysis revealed several valuable business insights:

- Customer spending patterns differ across demographic groups.
- Subscribers contribute significantly to business revenue.
- Certain products consistently receive higher customer ratings.
- Loyal customers account for a major share of purchases.
- Discount campaigns influence purchasing behavior for selected products.
- Shipping preferences are associated with different spending patterns.
- Customer segmentation helps identify high-value customer groups for targeted marketing.

9. Business Recommendations

Based on the analysis, the following recommendations are suggested:

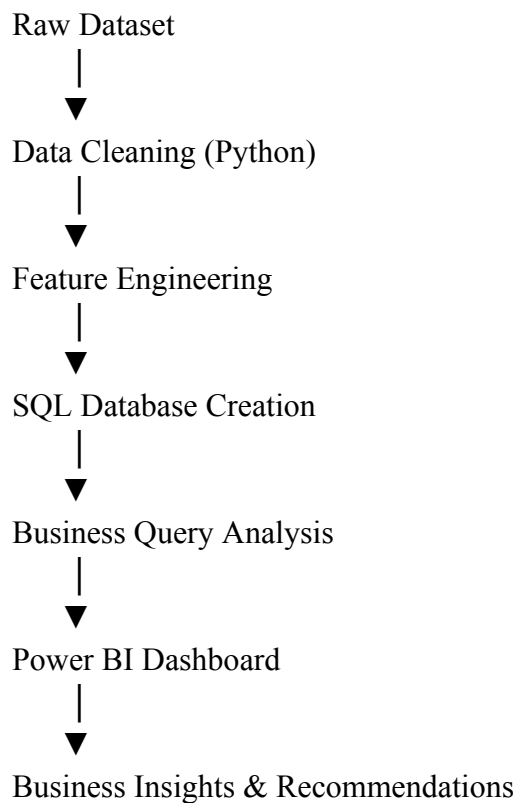
- Focus promotional campaigns on loyal and returning customers.
- Expand subscription programs to improve customer retention.
- Promote highly rated products through featured recommendations.
- Optimize inventory for top-selling products in each category.
- Personalize offers according to customer age groups and purchase history.
- Reduce unnecessary discounts on products with consistently high demand.
- Encourage premium shipping options for high-value customers.

10. Technologies Used

Technology	Purpose
Python	Data Cleaning & Feature Engineering
Pandas	Data Manipulation

NumPy	Numerical Operations
Matplotlib	Data Visualization
SQL Server (SSMS)	Business Query Analysis
SQL	Data Retrieval & Business Insights
Power BI	Dashboard Development

11. Project Workflow



12. Conclusion

This project demonstrates an end-to-end data analytics workflow by integrating Python, SQL, and Power BI to transform raw customer transaction data into meaningful business insights. The analysis highlights customer purchasing patterns, loyalty trends, product performance, and revenue drivers, enabling retailers to make informed decisions regarding marketing, inventory management, customer retention, and sales optimization. The project showcases practical skills in data cleaning, feature engineering, SQL querying, data visualization, and business intelligence, making it a strong portfolio piece for aspiring Data Analysts.